Skill-7

Input-

keathana@Chandu: ~/skill-6/S-7

GNU nano 8.7.1 process_monitor.c *

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <sys/types.h>
#include <string.h>
#include <limits.h>

void find_command(char *command)
{
    char *path = getenv("PATH");

    if (path == NULL)
    {
        printf("PATH variable not found.\n");
        return;
    }

    char *path_copy = strdup(path);
    char *dir = strtok(path_copy, ":");

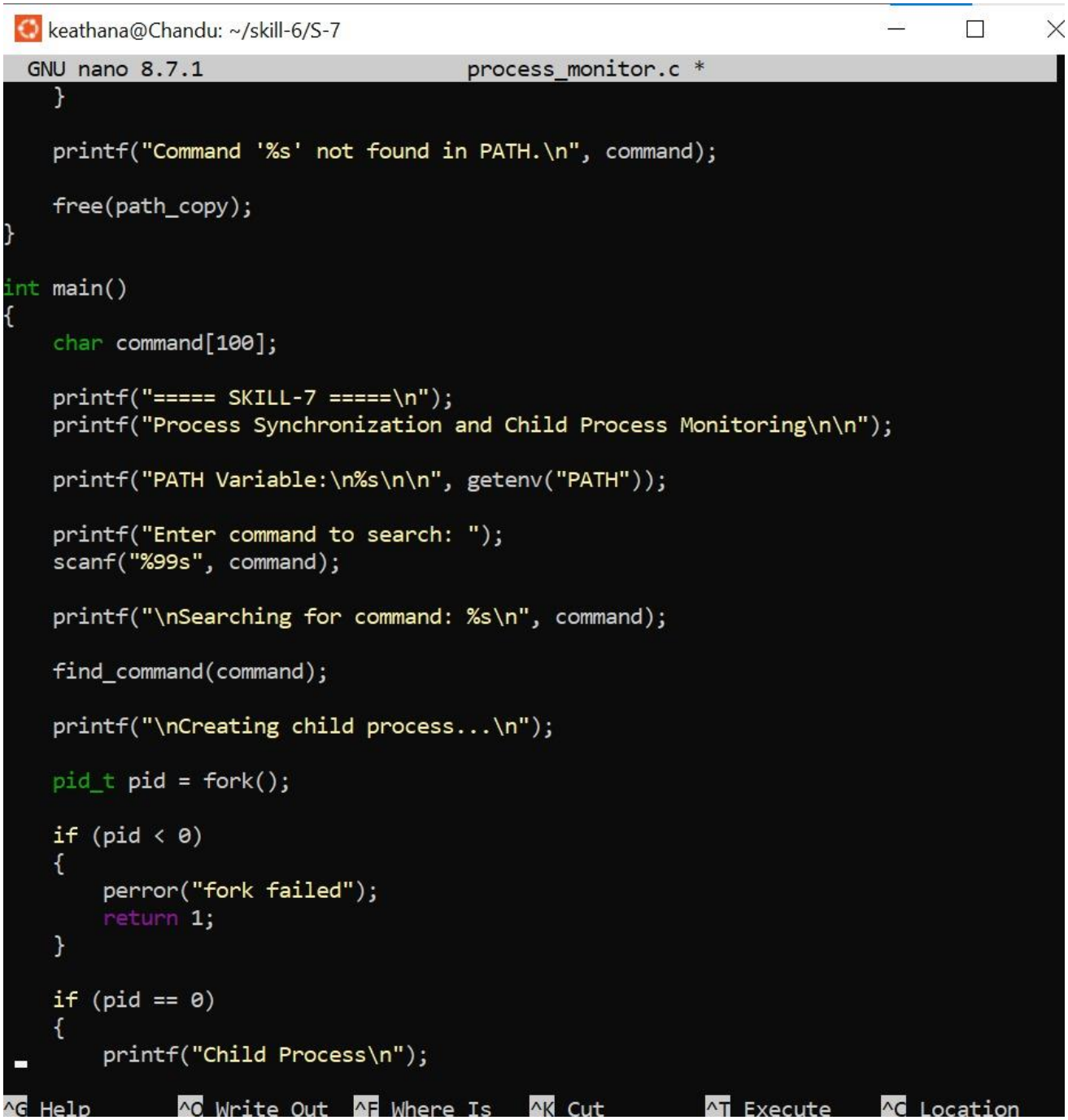
    while (dir != NULL)
    {
        char fullpath[PATH_MAX];

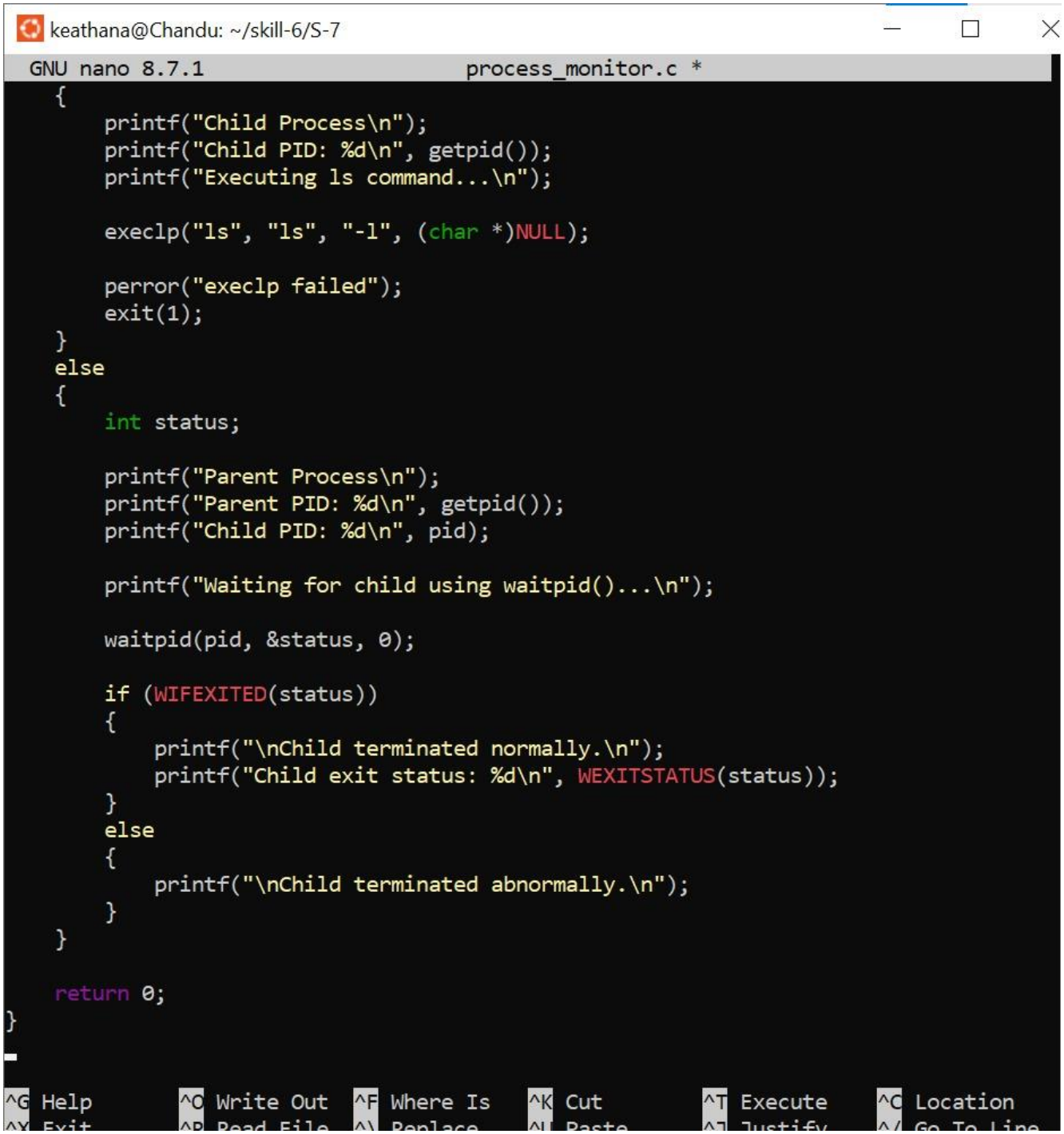
        snprintf(fullpath, sizeof(fullpath), "%s/%s", dir, command);

        if (access(fullpath, X_OK) == 0)
        {
            printf("Executable found: %s\n", fullpath);
            free(path_copy);
            return;
        }

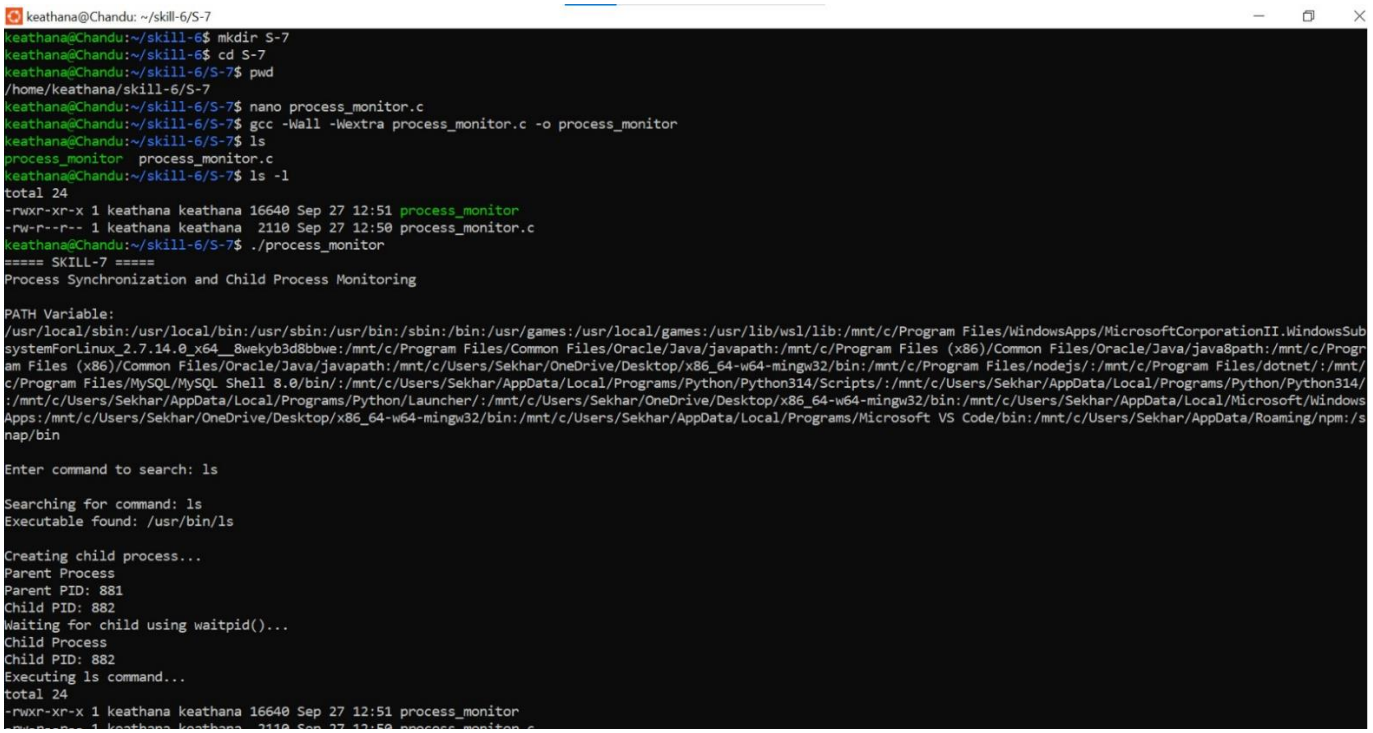
        dir = strtok(NULL, ":");
    }
```

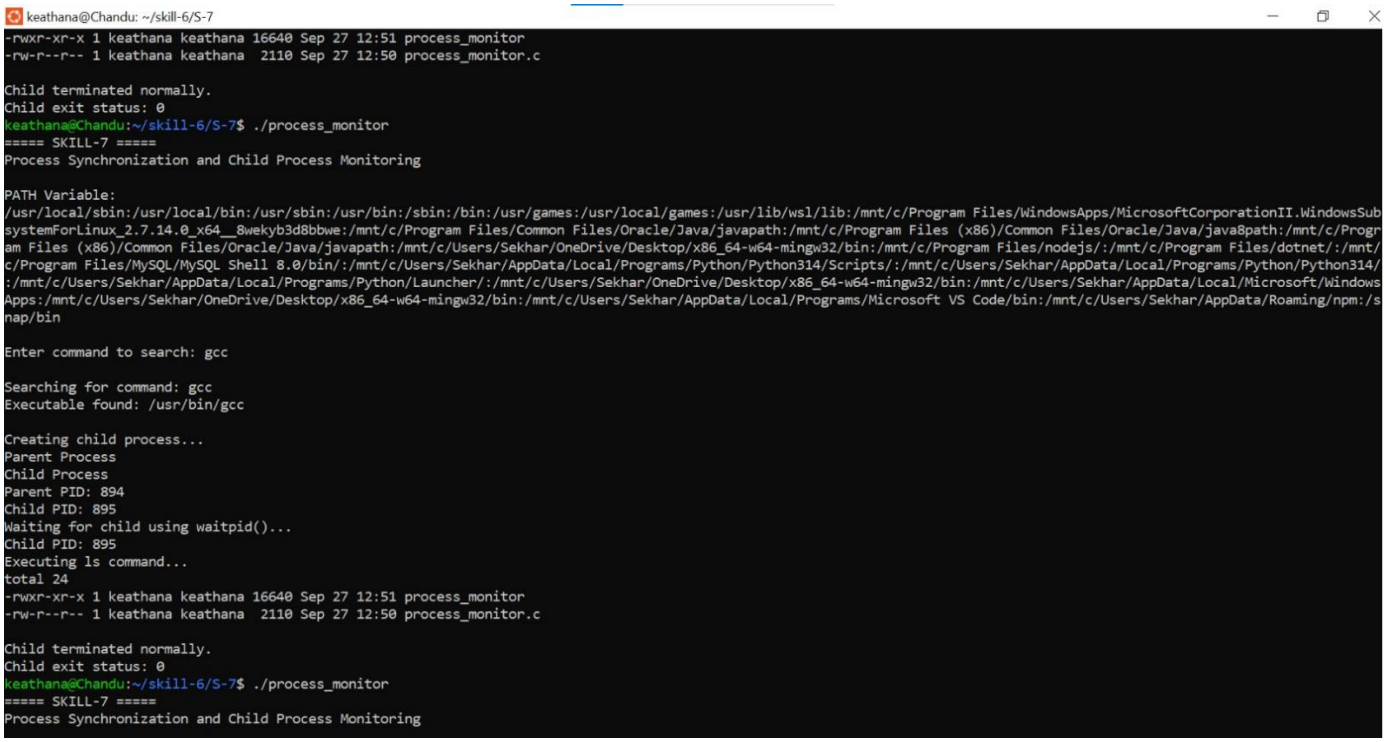
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location

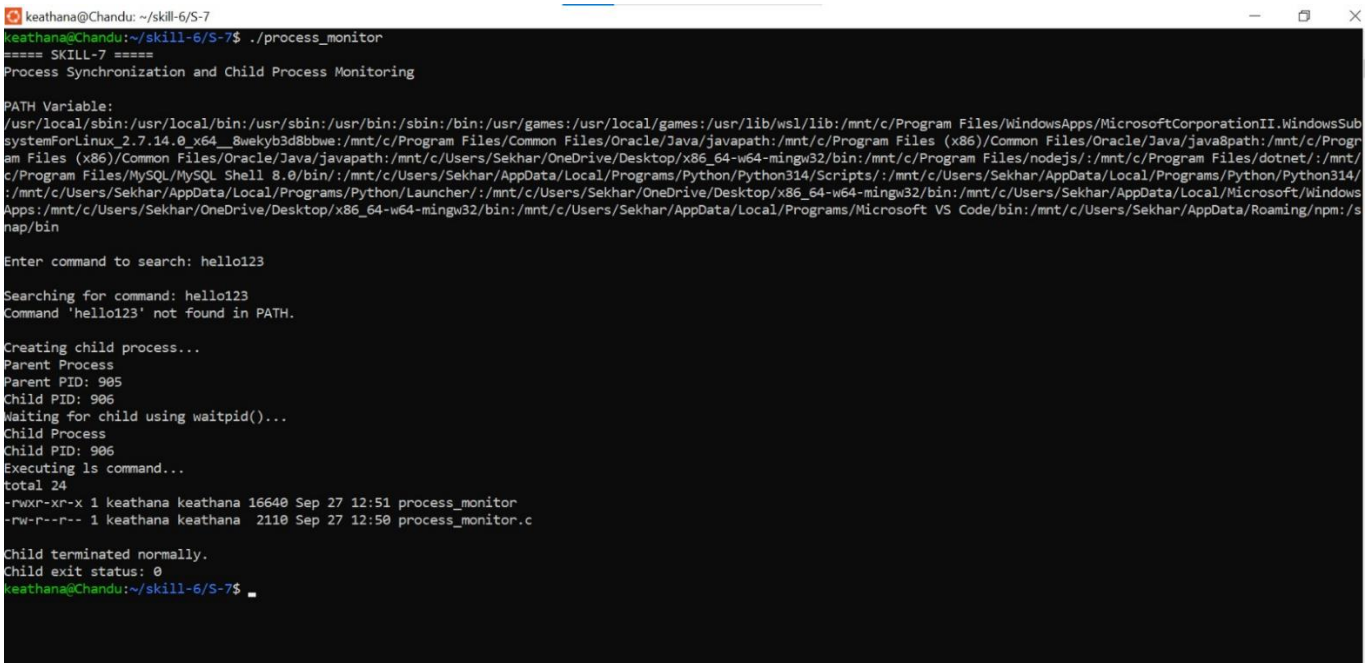

keathana@Chandu: ~/skill-6/S-7
GNU nano 8.7.1 process_monitor.c *
}
printf("Command '%s' not found in PATH.\n", command);
free(path_copy);
}
int main()
{
char command[100];
printf("===== SKILL-7 =====\n");
printf("Process Synchronization and Child Process Monitoring\n\n");
printf("PATH Variable:\n%s\n\n", getenv("PATH"));
printf("Enter command to search: ");
scanf("%99s", command);
printf("\nSearching for command: %s\n", command);
find_command(command);
printf("\nCreating child process...\n");
pid_t pid = fork();
if (pid < 0)
{
perror("fork failed");
return 1;
}
if (pid == 0)
{
printf("Child Process\n");
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location


keathana@Chandu: ~/skill-6/S-7
GNU nano 8.7.1 process_monitor.c *
{
printf("Child Process\n");
printf("Child PID: %d\n", getpid());
printf("Executing ls command...\n");
execlp("ls", "ls", "-l", (char *)NULL);
perror("execlp failed");
exit(1);
}
else
{
int status;
printf("Parent Process\n");
printf("Parent PID: %d\n", getpid());
printf("Child PID: %d\n", pid);
printf("Waiting for child using waitpid()...\n");
waitpid(pid, &status, 0);
if (WIFEXITED(status))
{
printf("\nChild terminated normally.\n");
printf("Child exit status: %d\n", WEXITSTATUS(status));
}
else
{
printf("\nChild terminated abnormally.\n");
}
}
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

OUTPUT:


keathana@Chandu: ~/skill-6/S-7
keathana@Chandu:~/skill-6$ mkdir S-7
keathana@Chandu:~/skill-6$ cd S-7
keathana@Chandu:~/skill-6/S-7$ pwd
/home/keathana/skill-6/S-7
keathana@Chandu:~/skill-6/S-7$ nano process_monitor.c
keathana@Chandu:~/skill-6/S-7$ gcc -Wall -Wextra process_monitor.c -o process_monitor
keathana@Chandu:~/skill-6/S-7$ ls
process_monitor process_monitor.c
keathana@Chandu:~/skill-6/S-7$ ls -l
total 24
-rwxr-xr-x 1 keathana keathana 16640 Sep 27 12:51 process_monitor
-rw-r--r-- 1 keathana keathana 2110 Sep 27 12:50 process_monitor.c
keathana@Chandu:~/skill-6/S-7$ ./process_monitor
===== SKILL-7 =====
Process Synchronization and Child Process Monitoring
PATH Variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.7.14.0_x64__8wekyb3d8bbwe:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8path:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Users/Sekhar/OneDrive/Desktop/x86_64-w64-mingw32/bin:/mnt/c/Program Files/nodejs/:/mnt/c/Program Files/dotnet/:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin/:/mnt/c/Users/Sekhar/AppData/Local/Programs/Python/Python314/Scripts/:/mnt/c/Users/Sekhar/AppData/Local/Programs/Python/Python314/:/mnt/c/Users/Sekhar/AppData/Local/Programs/Python/Launcher/:/mnt/c/Users/Sekhar/OneDrive/Desktop/x86_64-w64-mingw32/bin:/mnt/c/Users/Sekhar/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/Sekhar/OneDrive/Desktop/x86_64-w64-mingw32/bin:/mnt/c/Users/Sekhar/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/Sekhar/AppData/Roaming/npm:/snap/bin
Enter command to search: ls
Searching for command: ls
Executable found: /usr/bin/ls
Creating child process...
Parent Process
Parent PID: 881
Child PID: 882
Waiting for child using waitpid()...
Child Process
Child PID: 882
Executing ls command...
total 24
-rwxr-xr-x 1 keathana keathana 16640 Sep 27 12:51 process_monitor


keathana@Chandu: ~/skill-6/S-7
-rwxr-xr-x 1 keathana keathana 16640 Sep 27 12:51 process_monitor
-rw-r--r-- 1 keathana keathana 2110 Sep 27 12:50 process_monitor.c
Child terminated normally.
Child exit status: 0
keathana@Chandu:~/skill-6/S-7$ ./process_monitor
===== SKILL-7 =====
Process Synchronization and Child Process Monitoring
PATH Variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.7.14.0_x64__8wekyb3d8bbwe:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8path:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Users/Sekhar/OneDrive/Desktop/x86_64-w64-mingw32/bin:/mnt/c/Program Files/nodejs/:/mnt/c/Program Files/dotnet/:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin/:/mnt/c/Users/Sekhar/AppData/Local/Programs/Python/Python314/Scripts/:/mnt/c/Users/Sekhar/AppData/Local/Programs/Python/Python314/:/mnt/c/Users/Sekhar/AppData/Local/Programs/Python/Launcher/:/mnt/c/Users/Sekhar/OneDrive/Desktop/x86_64-w64-mingw32/bin:/mnt/c/Users/Sekhar/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/Sekhar/OneDrive/Desktop/x86_64-w64-mingw32/bin:/mnt/c/Users/Sekhar/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/Sekhar/AppData/Roaming/npm:/snap/bin
Enter command to search: gcc
Searching for command: gcc
Executable found: /usr/bin/gcc
Creating child process...
Parent Process
Child Process
Parent PID: 894
Child PID: 895
Waiting for child using waitpid()...
Child PID: 895
Executing ls command...
total 24
-rwxr-xr-x 1 keathana keathana 16640 Sep 27 12:51 process_monitor
-rw-r--r-- 1 keathana keathana 2110 Sep 27 12:50 process_monitor.c
Child terminated normally.
Child exit status: 0
keathana@Chandu:~/skill-6/S-7$ ./process_monitor
===== SKILL-7 =====
Process Synchronization and Child Process Monitoring


keathana@Chandu: ~/skill-6/S-7
keathana@Chandu:~/skill-6/S-7$ ./process_monitor
===== SKILL-7 =====
Process Synchronization and Child Process Monitoring
PATH Variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.7.14.0_x64__8wekyb3d8bbwe:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8path:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Users/Sekhar/OneDrive/Desktop/x86_64-w64-mingw32/bin:/mnt/c/Program Files/nodejs/:/mnt/c/Program Files/dotnet/:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin/:/mnt/c/Users/Sekhar/AppData/Local/Programs/Python/Python314/Scripts/:/mnt/c/Users/Sekhar/AppData/Local/Programs/Python/Python314/:/mnt/c/Users/Sekhar/AppData/Local/Programs/Python/Launcher/:/mnt/c/Users/Sekhar/OneDrive/Desktop/x86_64-w64-mingw32/bin:/mnt/c/Users/Sekhar/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/Sekhar/OneDrive/Desktop/x86_64-w64-mingw32/bin:/mnt/c/Users/Sekhar/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/Sekhar/AppData/Roaming/npm:/snap/bin
Enter command to search: hello123
Searching for command: hello123
Command 'hello123' not found in PATH.
Creating child process...
Parent Process
Parent PID: 905
Child PID: 906
Waiting for child using waitpid()...
Child Process
Child PID: 906
Executing ls command...
total 24
-rwxr-xr-x 1 keathana keathana 16640 Sep 27 12:51 process_monitor
-rw-r--r-- 1 keathana keathana 2110 Sep 27 12:50 process_monitor.c
Child terminated normally.
Child exit status: 0
keathana@Chandu:~/skill-6/S-7$